

PART 3: COMBINATIONAL LOGIC

MULTIPLEXERS

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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MIDTERM TOPICS

- November 12th (Saturday) 8:30 am to 11:30 am
 - Number Systems
 - Boolean Algebra
 - K-maps and logic simplification
 - Logic gate implementations,
 - Combinational Logic (Binary Adder/Subtractor, Decimal Adder; Binary Multiplier; Comparator; Decoder, Encoder; Multiplexer)

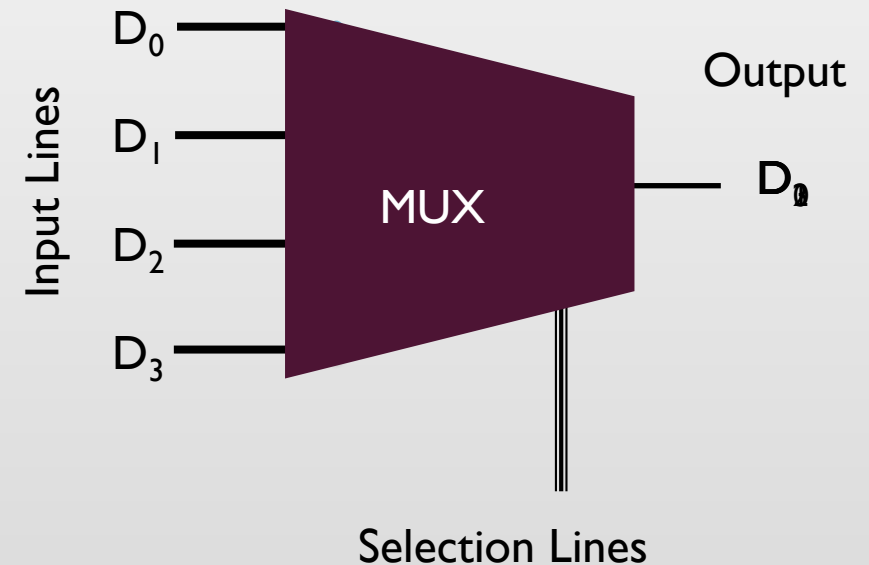
MONDAY MINI-QUIZ



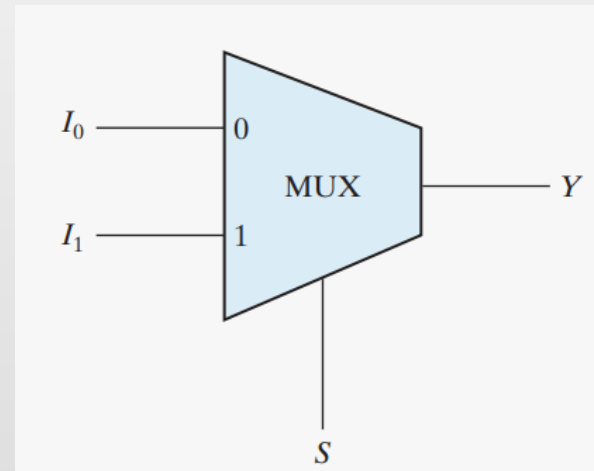
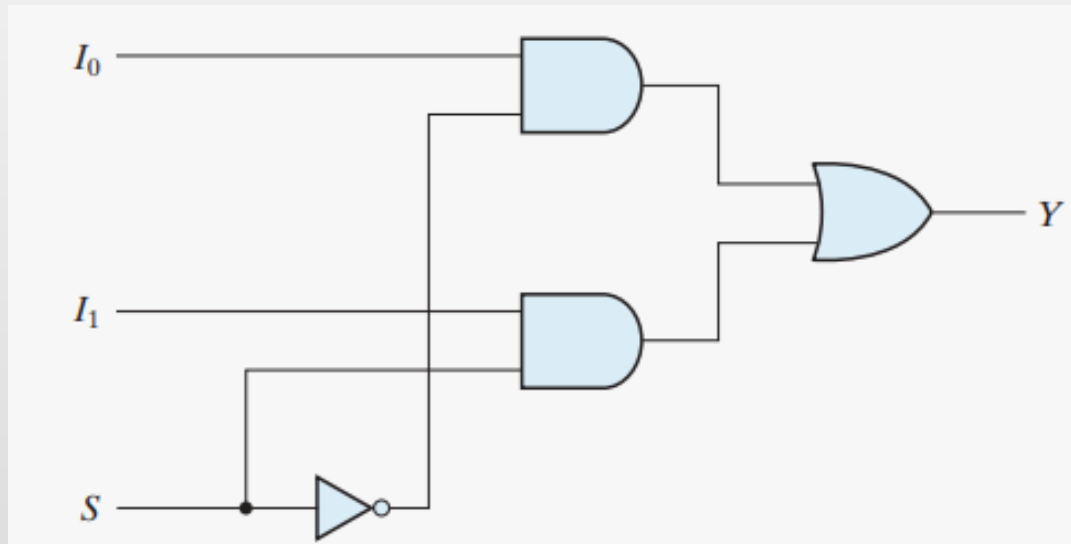
- Mini-quiz (15 to 20 minute) next week on Monday 24th, counts for 2% of the grade
- Decoders, encoders and multiplexers.

MULTIPLEXER

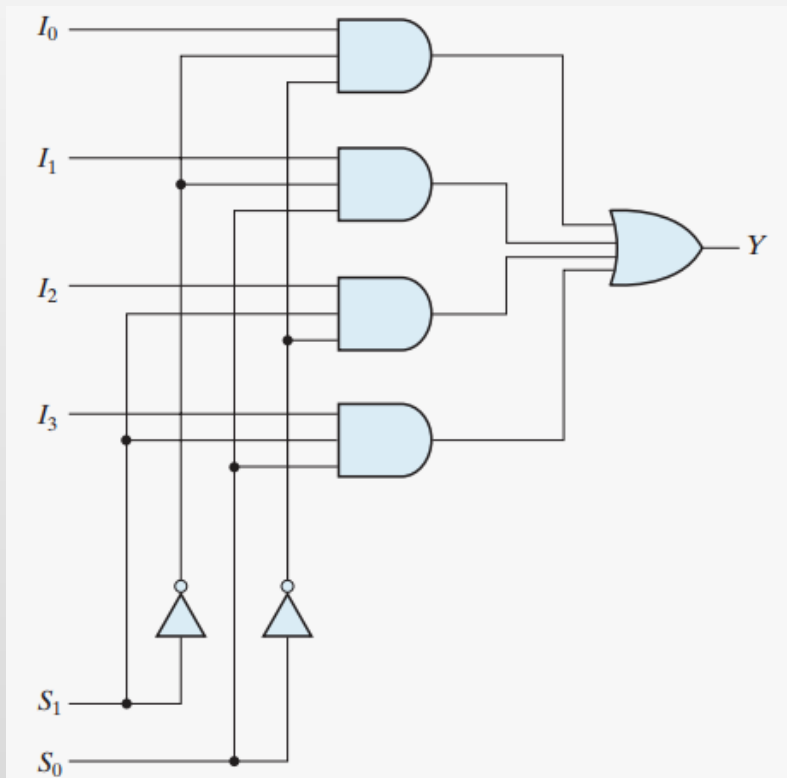
- A multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line.
- The selection of particular input line is controlled by a set of selection lines.
- Normally there are 2^n input lines, and n selection lines whose bit combinations determine which input is selected.



TWO-TO-ONE LINE MUX



FOUR-TO-ONE LINE MUX



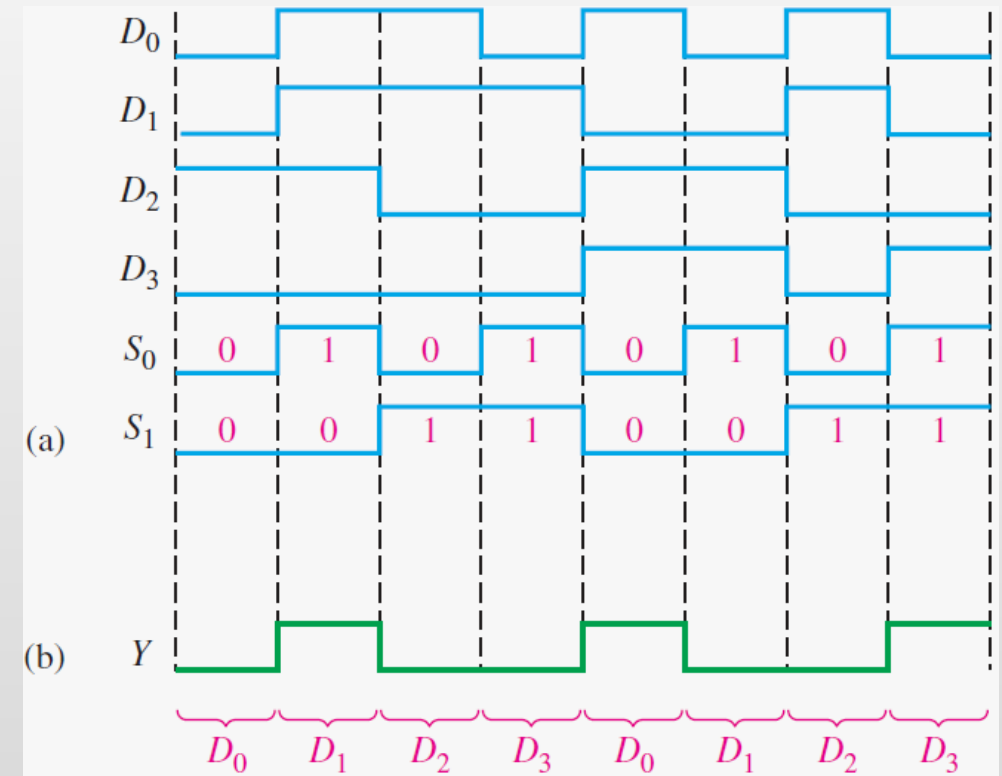
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

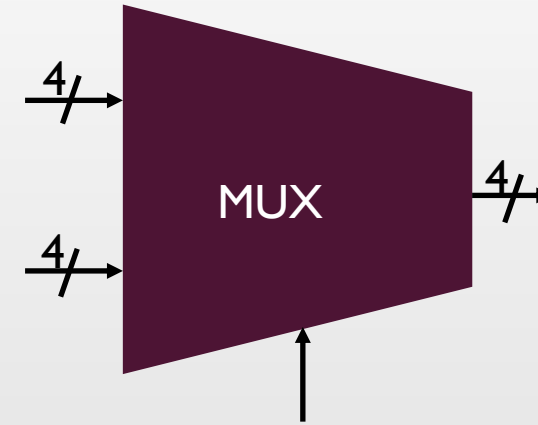
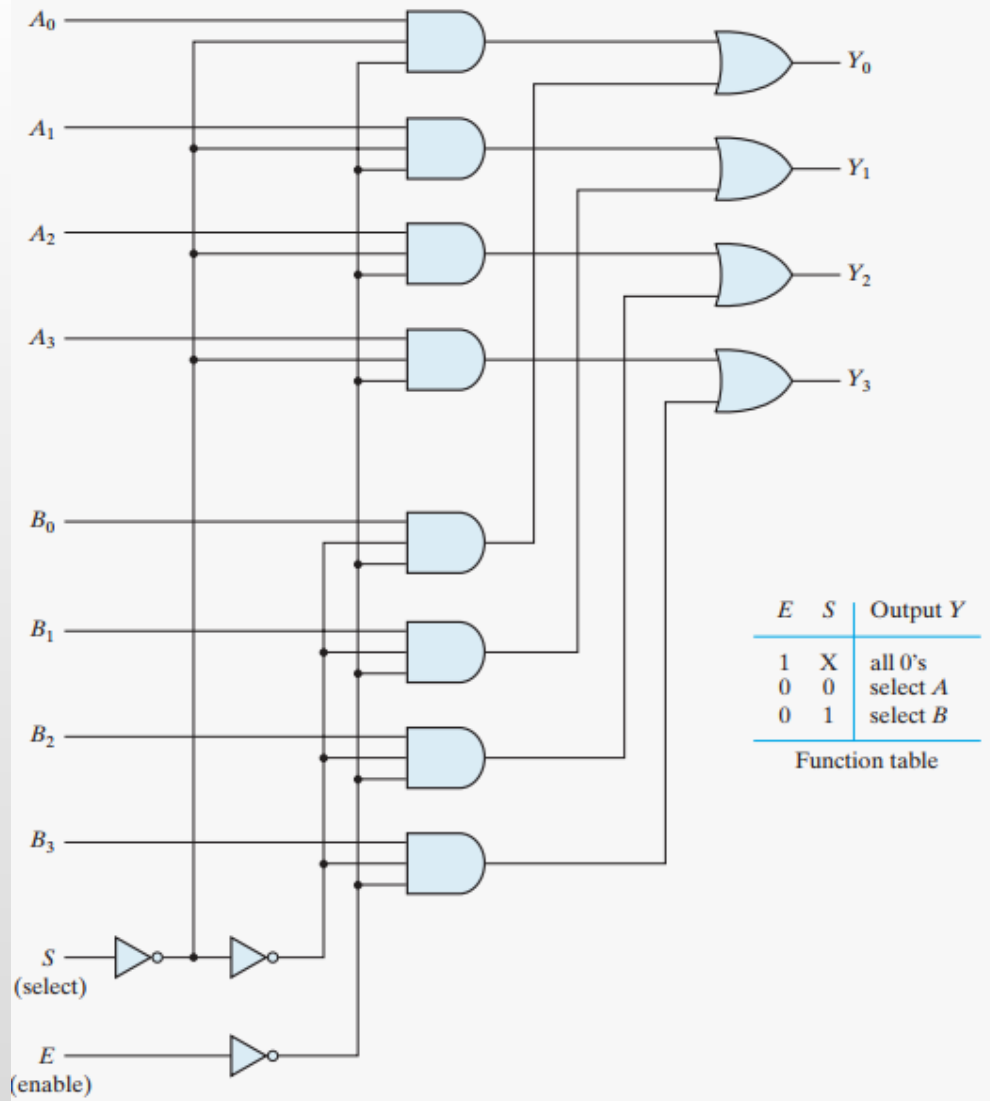
As in decoders, multiplexers may have an enable input to control the operation of the unit.

FOUR-TO-ONE LINE MUX EXAMPLE

- The data-input and data-select waveforms in the circuit are applied to a four-to-one line mux multiplexer. Determine the output waveform in relation to the inputs.

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



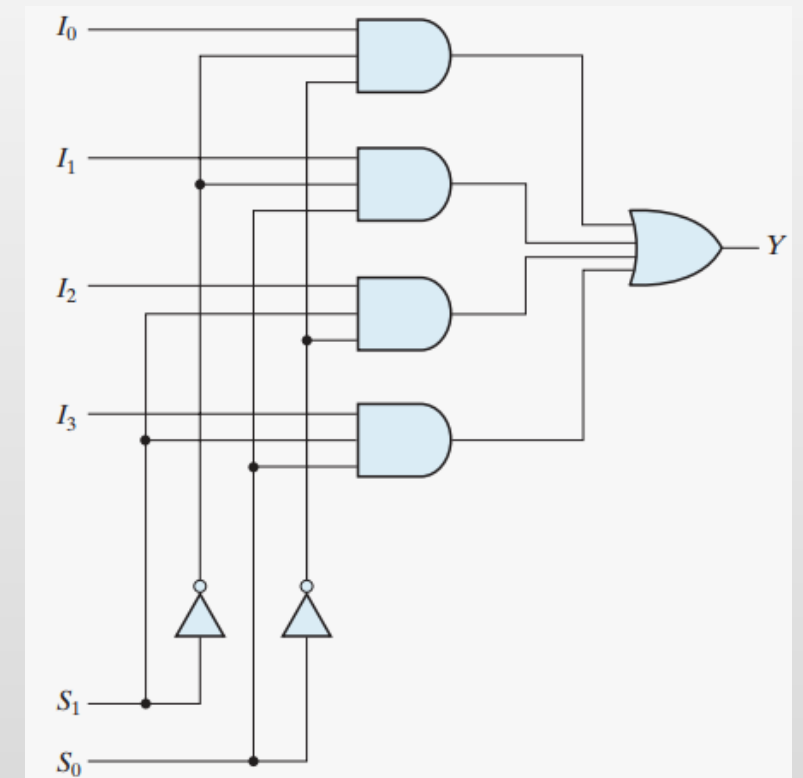


QUADRUPL 2-TO-1 MUX

Multiplexer circuits can be combined with common selection inputs to provide multiple-bit selection logic.

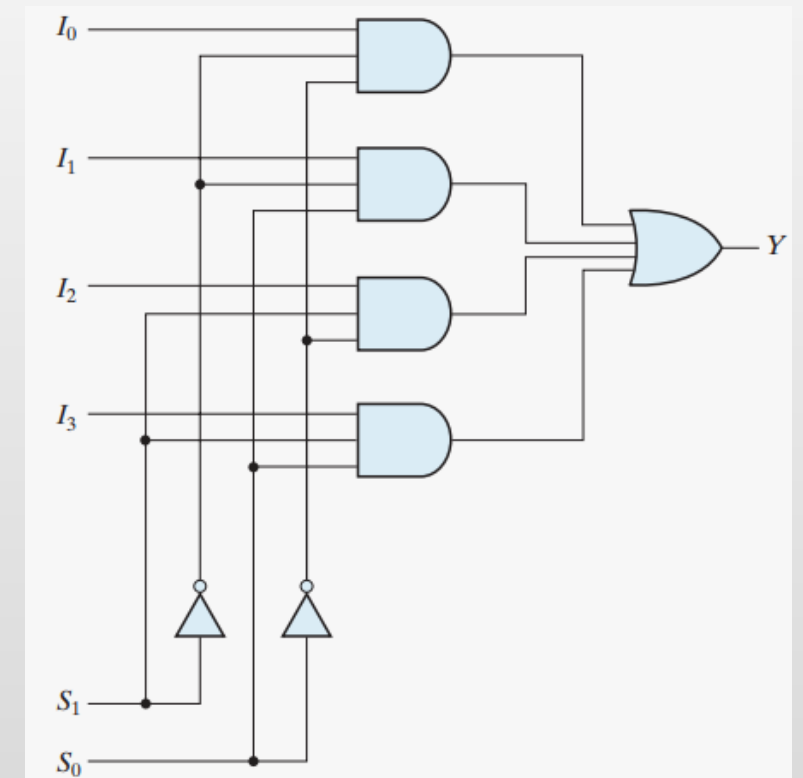
BOOLEAN FUNCTION IMPLEMENTATION

- Multiplexer is essentially a decoder that includes an OR gate within the unit.
- The minterms of a function are generated in a multiplexer by the circuit associated with the selection inputs.
- The individual minterms can be selected by the data inputs.



BOOLEAN FUNCTION IMPLEMENTATION

- Boolean function of n -variables can be implemented with multiplexer having $(n - 1)$ selection lines.
- The first $n - 1$ variables of the function are connected to the selection inputs of the multiplexer.
- The remaining single variable of the function is used for the data inputs. If the single variable is denoted by z , each data input of the multiplexer will be $z, z', 1$, or 0 .



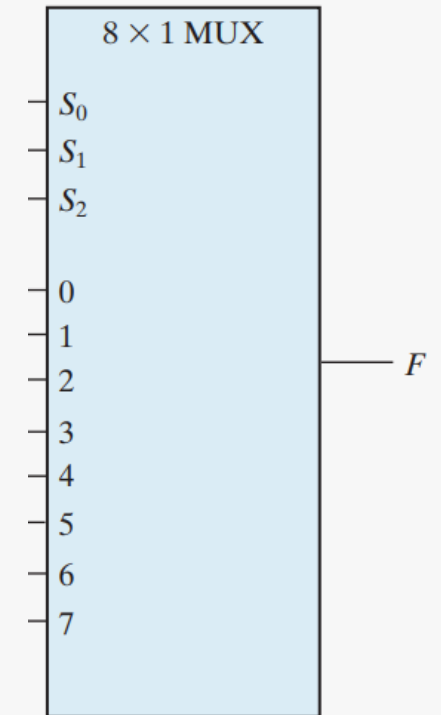
BOOLEAN FUNCTION IMPLEMENTATION

■ Example

Using 8x1 MUX to implement:

$$F(A, B, C, D) = \sum(1, 3, 4, 11, 12, 13, 14, 15)$$

A	B	C	D	F	
0	0	0	0	0	$F = D$
0	0	0	1	1	
0	0	1	0	0	$F = D$
0	0	1	1	1	
0	1	0	0	1	$F = D'$
0	1	0	1	0	
0	1	1	0	0	$F = 0$
0	1	1	1	0	
1	0	0	0	0	$F = 0$
1	0	0	1	0	
1	0	1	0	0	$F = D$
1	0	1	1	1	
1	1	0	0	1	$F = 1$
1	1	0	1	1	
1	1	1	0	1	$F = 1$
1	1	1	1	1	



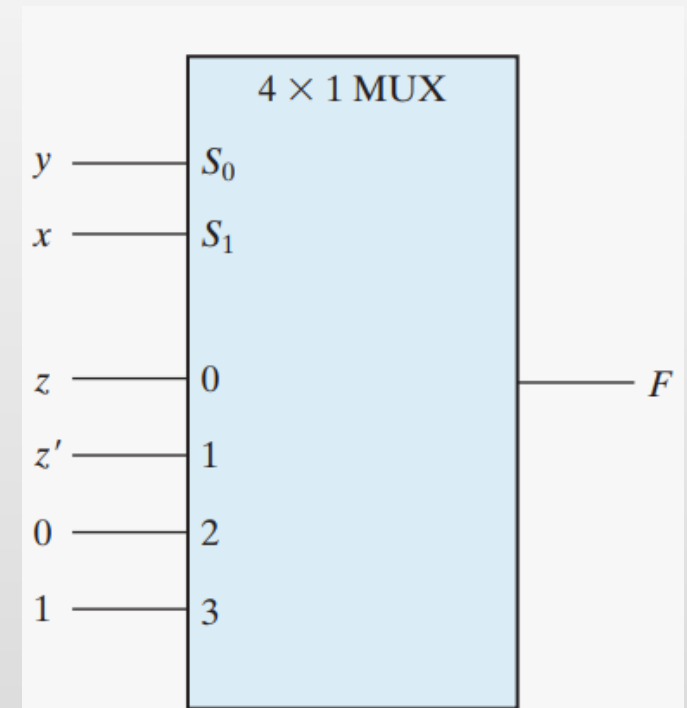
BOOLEAN FUNCTION IMPLEMENTATION

- **Exercise:** Implement the following Boolean function using a multiplexer

$$F(x, y, z) = \sum(1, 2, 6, 7)$$

x	y	z	F	
0	0	0	0	$F = z$
0	0	1	1	
0	1	0	1	$F = z'$
0	1	1	0	
1	0	0	0	$F = 0$
1	0	1	0	
1	1	0	1	$F = 1$
1	1	1	1	

(a) Truth table

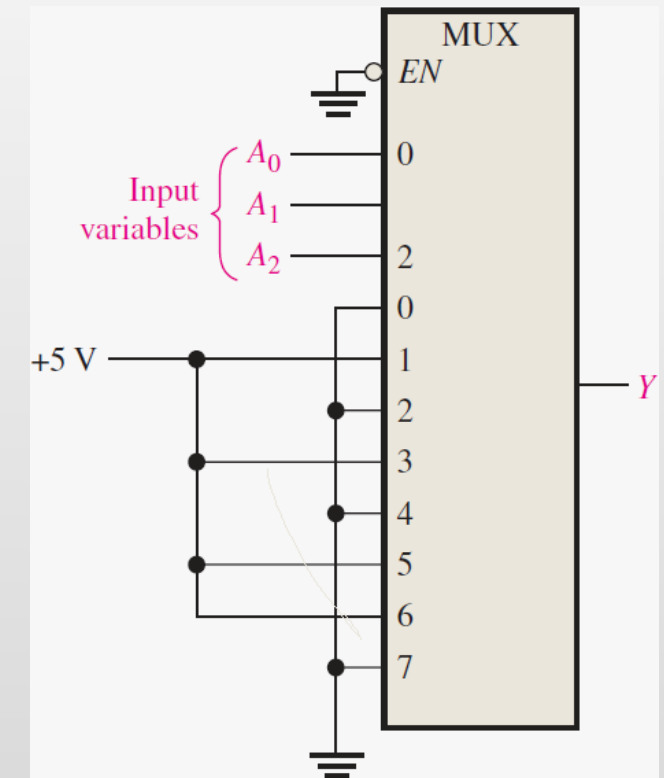


(b) Multiplexer implementation

BOOLEAN FUNCTION IMPLEMENTATION

- Implement the logic function specified in the table using a MUX.
- The implementation of this function would require four 3-input AND gates, one 4-input OR gate and three inverters unless the expression can be simplified.

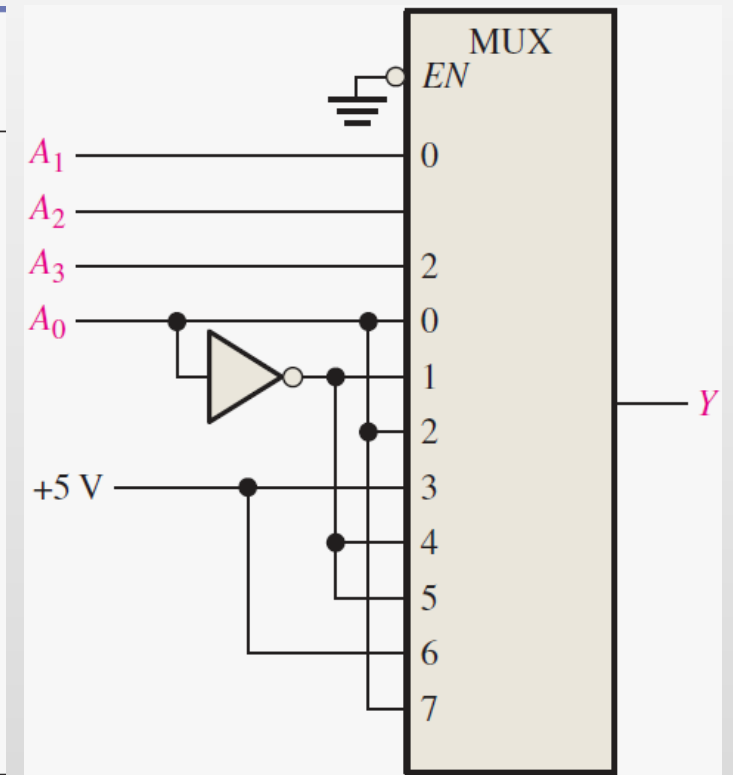
Inputs			Output
A_2	A_1	A_0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



BOOLEAN FUNCTION IMPLEMENTATION

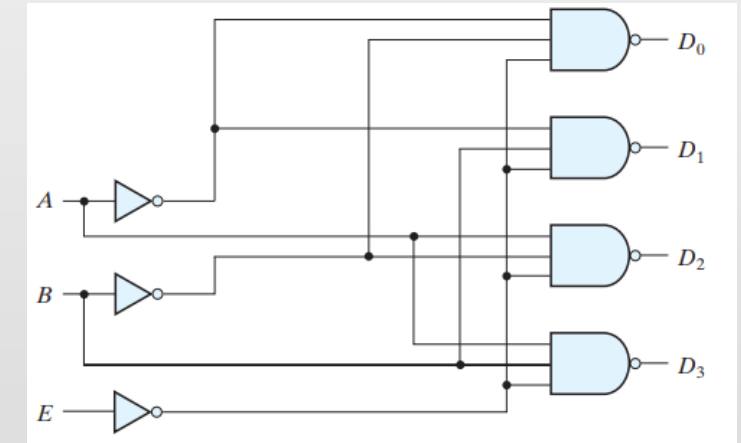
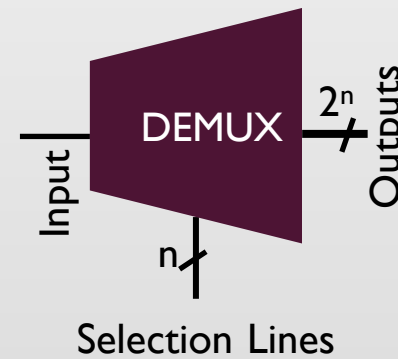
- Implement the logic function specified in the table using a MUX.

Decimal Digit	Inputs				Output Y
	A ₃	A ₂	A ₁	A ₀	
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	1
11	1	0	1	1	0
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

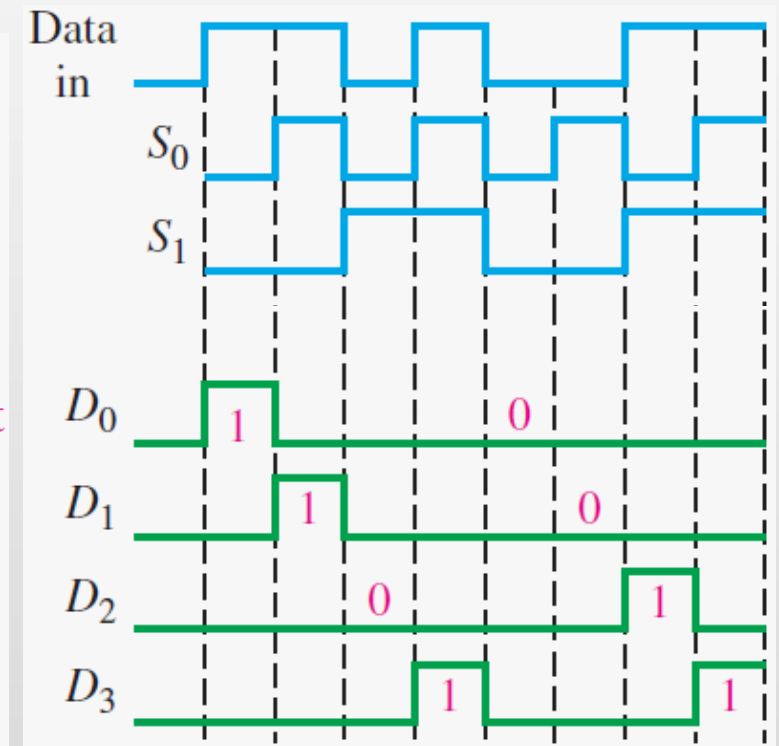
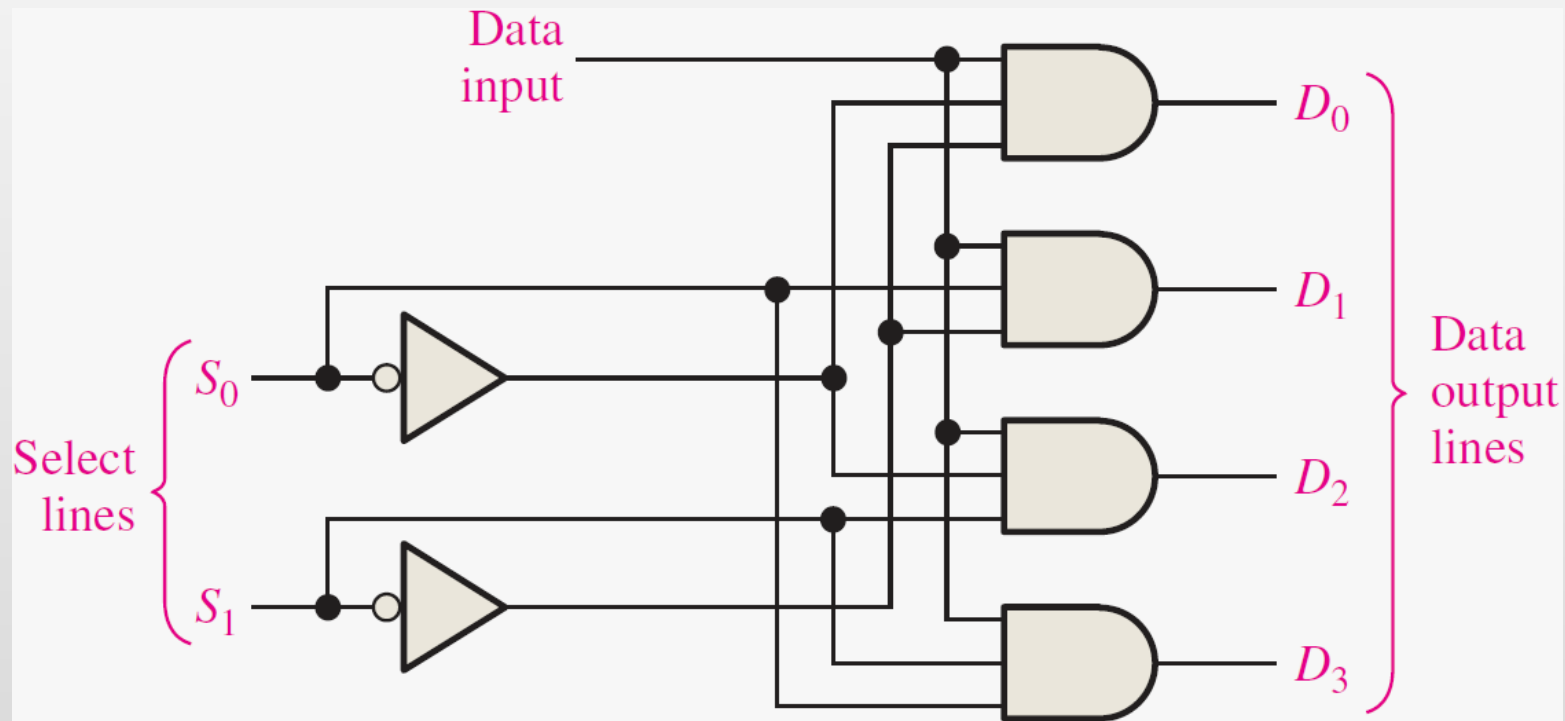


DEMULTIPLEXER

- A circuit that receives information from a single line and directs it to one of 2^n possible output lines. The selection of specific output is controlled the bit combination of selection lines.
- 2x4 decoder with Enable can function as one-to-four line demultiplexer when E is taken as data input line and A and B are taken as selection inputs.
- Decoder with enable input is referred to as decoder-demultiplexer.

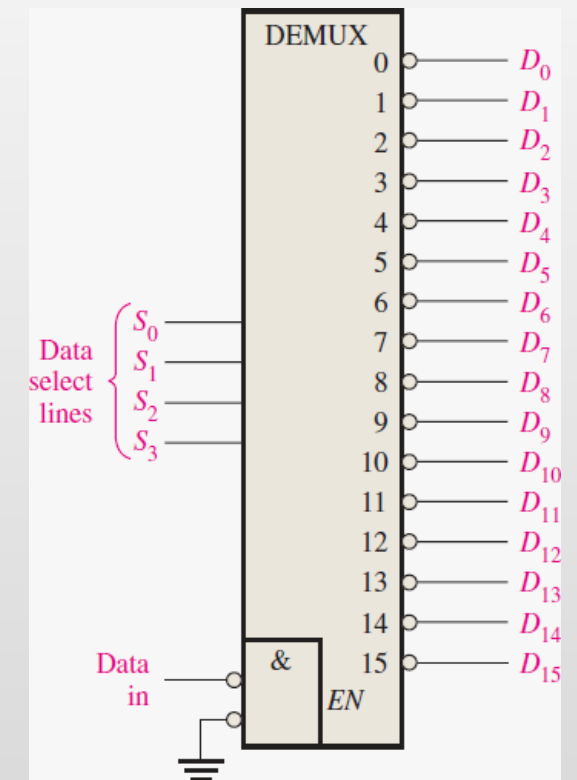


I-TO-4 DEMULTIPLEXER



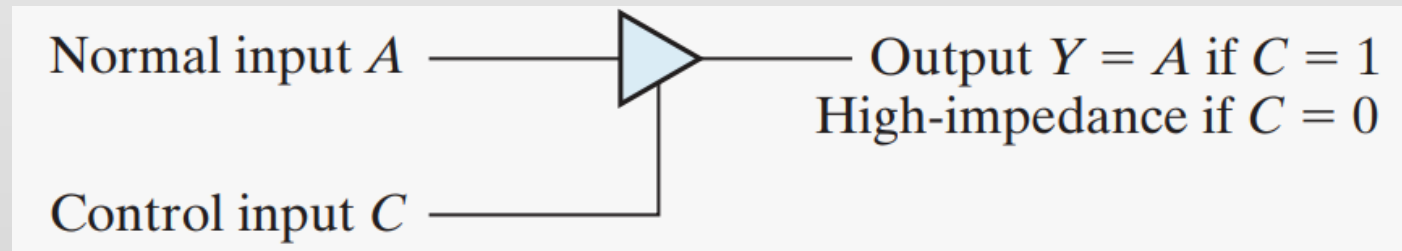
4-TO-16 LINE DECODER AS A DEMULTIPLEXER

- In demultiplexer applications, the input lines are used as the data-select lines. One of the chip select inputs is used as the data input line, with the other chip select input held LOW to enable the internal negative-AND gate at the bottom of the diagram.



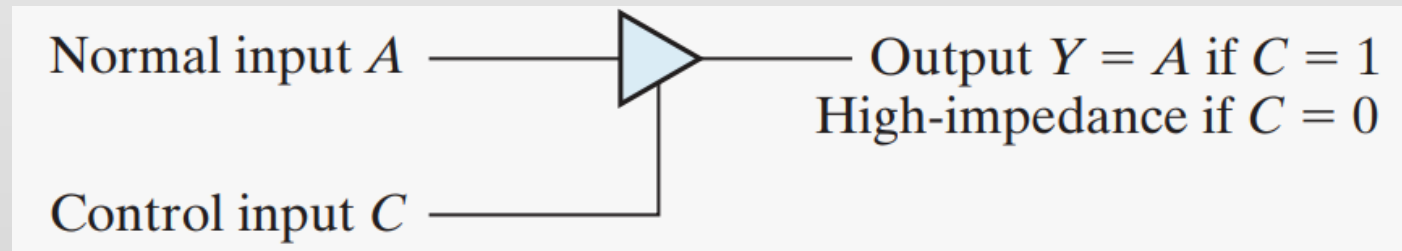
THREE-STATE GATES

- A multiplexer can be constructed with three state gates – digital circuits that exhibit three states. The one most commonly used is the buffer gate.
- Two of the states are signals equivalent to logic 1 and logic 0 as in a conventional gate.

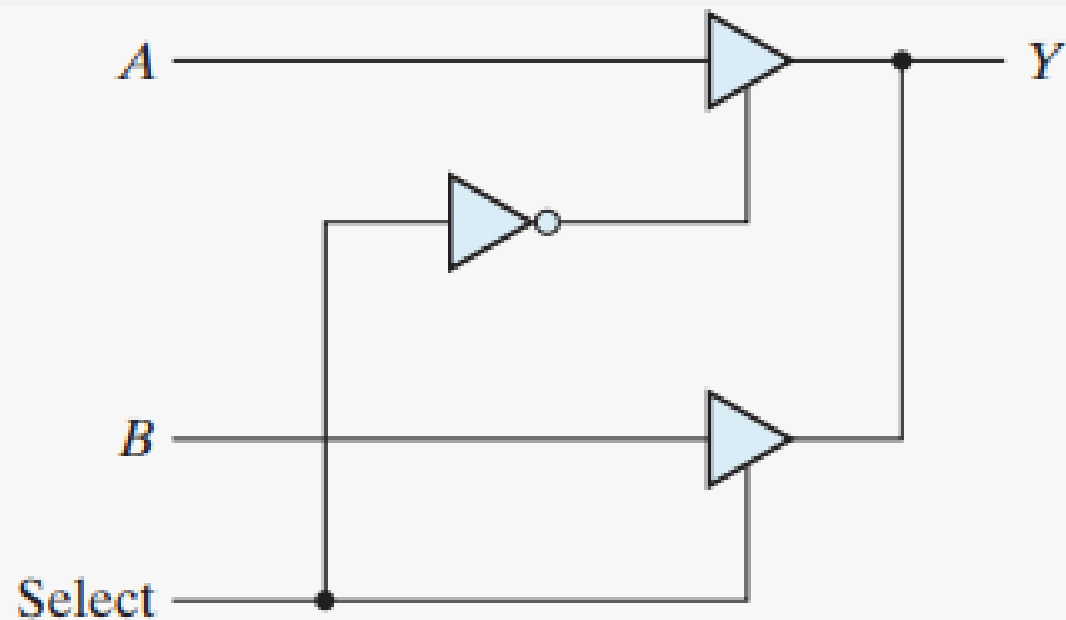


THREE-STATE GATES

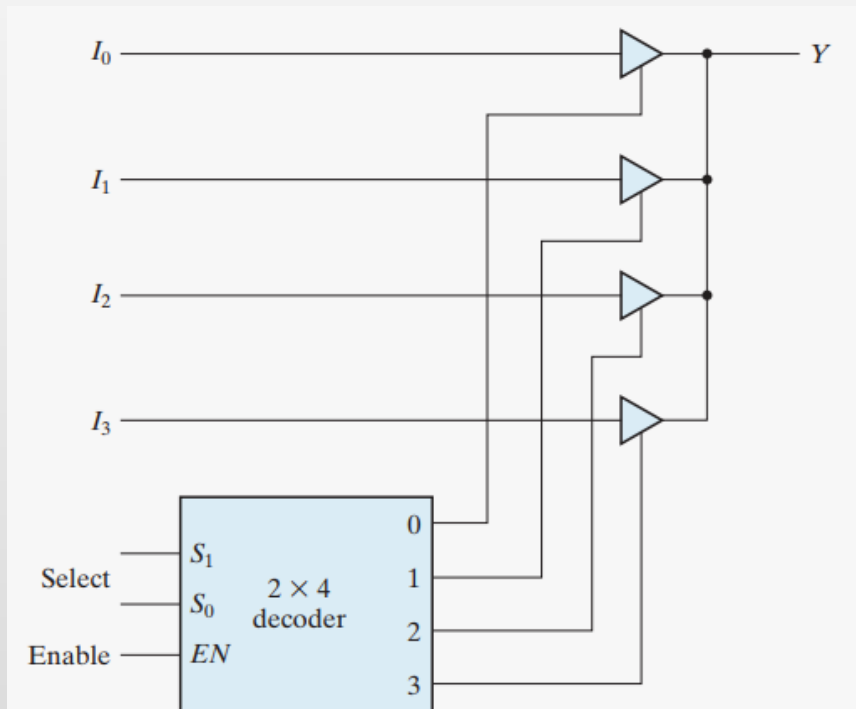
- The third state is a **high-impedance state** in which:
 1. The logic behaves like an open circuit, which means that the output appears to be disconnected.
 2. The circuit has no logic significance.
 3. The circuit connected to the output of the three-state gate is not affected by the inputs to the gate.



THREE-STATE GATES



(a) 2-to-1-line mux



(b) 4-to-1-line mux